

H524 Week 4 Assignment: Confidence Intervals

Student Version

Introduction to Biostatistics

Fall 2025

Assignment Information:

- **Total Points:** 110 points
- **Time Limit:** 180 minutes (3 hours)
- **Attempts:** 1
- **Question Types:** 16 Multiple Choice (80 pts), 2 Essay/Fill-in (30 pts)

Part 1: CI Fundamentals

Question 1: CI Interpretation (5 points)

A study reports a 95% confidence interval for mean blood pressure reduction as (8.2, 14.6) mmHg. What is the CORRECT interpretation?

- a) We are 95% confident that the true population mean blood pressure reduction is between 8.2 and 14.6 mmHg
- b) There is a 95% probability that the true mean is between 8.2 and 14.6 mmHg
- c) 95% of patients will have blood pressure reductions between 8.2 and 14.6 mmHg
- d) The sample mean has a 95% chance of being between 8.2 and 14.6 mmHg

Question 2: Critical Value Selection (5 points)

You want to construct a 95% confidence interval for a population mean. The sample size is $n = 15$, and the population standard deviation is unknown. Which critical value should you use?

- a) $z = 1.96$
- b) $t(14) = 2.145$
- c) $t(15) = 2.131$
- d) $z = 2.576$

Question 3: Factors Affecting CI Width (5 points)

If you want a NARROWER confidence interval for a population mean, which of the following actions would achieve this?

- a) Increase the confidence level from 95% to 99%
- b) Decrease the sample size
- c) Increase the sample size
- d) Use a larger standard deviation

Question 4: Standard Error (5 points)

A study has a sample standard deviation $s = 12$ and sample size $n = 36$. What is the standard error of the mean?

- a) $SE = 2$
- b) $SE = 3$
- c) $SE = 6$
- d) $SE = 12$

Part 2: CI for Means**Question 5: Manual CI Calculation (15 points)**

A clinical trial measures pain reduction scores for 25 patients receiving a new treatment. The results are:

- Sample mean: $\bar{x} = 7.8$ points
- Sample standard deviation: $s = 3.2$ points
- Sample size: $n = 25$

Calculate a 95% confidence interval for the true mean pain reduction. Show all work including:

- a) Which distribution to use (z or t) and why (3 points)
- b) The appropriate critical value (3 points)
- c) The standard error calculation (3 points)
- d) The margin of error calculation (3 points)
- e) The final confidence interval with interpretation (3 points)

Hint: For a 95% CI with $df=24$, the critical t -value is approximately 2.064

Question 6: Confidence Level Comparison (5 points)

Using the same data from the Manual CI question above, if we constructed a 99% confidence interval instead of 95%, what would happen to the interval width?

- a) The interval would be narrower because we are more confident
- b) The interval would be wider because higher confidence requires a larger margin of error
- c) The interval width would stay exactly the same
- d) The interval would be narrower because the critical value decreases

Part 3: CI for Proportions**Question 7: Proportion CI Conditions (5 points)**

When constructing a confidence interval for a population proportion using the normal approximation (Wald method), which condition must be satisfied?

- a) $n \geq 30$
- b) Both $np \geq 5$ and $n(1 - p) \geq 5$
- c) The population must be normally distributed
- d) The sample proportion must be greater than 0.5

Question 8: Proportion CI Calculation (5 points)

In a survey of 200 patients, 150 reported satisfaction with their treatment. What is the sample proportion ($\hat{p}$)?

- a) $\hat{p} = 0.50$
- b) $\hat{p} = 0.67$
- c) $\hat{p} = 0.75$
- d) $\hat{p} = 0.80$

Question 9: Standard Error for Proportion (5 points)

Using the data from the proportion calculation question ($\hat{p} = 0.75$, $n = 200$), what is the standard error for the proportion? Use $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

- a) $SE \approx 0.031$
- b) $SE \approx 0.043$
- c) $SE \approx 0.075$
- d) $SE \approx 0.125$

Question 10: Vaccine Efficacy CI (5 points)

A vaccine trial reports a 95% CI for vaccine efficacy as (72%, 89%). What can we conclude?

- a) We are 95% confident the true efficacy is between 72% and 89%, suggesting the vaccine is effective since the entire interval is above 0%
- b) 72% of people are protected at minimum, 89% at maximum
- c) The vaccine has a 95% chance of working
- d) We cannot determine effectiveness because the interval is too wide

Part 4: Sample Size Determination**Question 11: Sample Size Concepts (5 points)**

You want to estimate the mean cholesterol level in a population with a margin of error of 5 mg/dL at 95% confidence. If you quadruple the sample size, what happens to the margin of error?

- a) The margin of error is cut in half
- b) The margin of error is cut to one-quarter
- c) The margin of error doubles
- d) The margin of error stays the same

Question 12: Sample Size Calculation (5 points)

A researcher wants to estimate a population proportion with a 95% CI and margin of error of 0.05. Using the conservative estimate ($\hat{p} = 0.5$), what sample size is needed?

Formula: $n = \frac{z^2 \cdot \hat{p}(1-\hat{p})}{E^2}$, where $z = 1.96$ for 95% CI

- a) $n = 196$
- b) $n = 271$
- c) $n = 385$
- d) $n = 577$

Part 5: Assumptions and Checking**Question 13: Normality Assessment (5 points)**

You have a sample of $n = 12$ observations and want to construct a t -based confidence interval for the mean. Which of the following is the BEST way to assess whether the normality assumption is reasonable?

- a) Check that $n \geq 30$, which automatically ensures normality

- b) Create a histogram and Q-Q plot to visually assess normality, and consider the Shapiro-Wilk test
- c) Calculate the mean and median; if they're close, the data is normal
- d) Normality is not required for t -based intervals

Question 14: Large vs Small Sample (5 points)

When can you safely use a z -based confidence interval instead of a t -based confidence interval for a population mean?

- a) When the population standard deviation (σ) is known, regardless of sample size
- b) Only when $n \geq 30$
- c) When the sample size is less than 30
- d) Never; always use t -distribution for real-world data

Question 15: CI for Hypothesis Testing (5 points)

A 95% confidence interval for the difference in mean blood pressure between two treatments is $(-2.3, 8.7)$ mmHg. What can you conclude at the $\alpha = 0.05$ significance level?

- a) There is a significant difference between treatments because the interval is positive
- b) There is NOT a significant difference between treatments because the interval contains 0
- c) Treatment 1 is better than Treatment 2
- d) We need more data to make any conclusion

Part 6: R Implementation**Question 16: CI with R (15 points)**

A clinical trial measures vitamin D levels (ng/mL) for 30 patients after supplementation:

```
vitamin_d <- c(32, 38, 29, 35, 31, 40, 27, 34, 36, 30,
               28, 37, 33, 35, 31, 34, 36, 29, 33, 37,
               30, 35, 34, 31, 39, 33, 36, 29, 34, 35)
```

Using R, complete the following tasks and paste your code and results:

- a) Calculate the sample mean, standard deviation, and sample size (3 points)
- b) Create a histogram to assess normality (2 points)
- c) Use `t.test()` to calculate a 95% confidence interval for the mean vitamin D level (5 points)
- d) Interpret the confidence interval in context (3 points)
- e) What would you conclude about whether the mean vitamin D level is above 30 ng/mL? (2 points)

a) Descriptive statistics (3 points):

```

vitamin_d <- c(32, 38, 29, 35, 31, 40, 27, 34, 36, 30,
              28, 37, 33, 35, 31, 34, 36, 29, 33, 37,
              30, 35, 34, 31, 39, 33, 36, 29, 34, 35)

# Sample statistics
mean(vitamin_d)
# [1] 33.37

sd(vitamin_d)
# [1] 3.35

length(vitamin_d)
# [1] 30

```

b) Histogram for normality (2 points):

```

hist(vitamin_d,
     main = "Histogram of Vitamin D Levels",
     xlab = "Vitamin D (ng/mL)",
     col = "lightblue",
     breaks = 10)

```

The histogram should show approximately normal distribution.

c) 95% CI using t.test() (5 points):

```

t.test(vitamin_d, conf.level = 0.95)

# Output:
# One Sample t-test
#
# data:  vitamin_d
# t = 54.595, df = 29, p-value < 2.2e-16
# alternative hypothesis: true mean is not equal to 0
# 95 percent confidence interval:
#  32.11669 34.61664
# sample estimates:
# mean of x
#  33.36667

```

d) Interpretation (3 points):

We are 95% confident that the true mean vitamin D level for all patients receiving this supplementation is between 32.12 and 34.62 ng/mL.

e) Conclusion about 30 ng/mL (2 points):

Since the entire 95% confidence interval (32.12, 34.62) is above 30 ng/mL, we can conclude with 95% confidence that the true mean vitamin D level is greater than 30 ng/mL. This suggests the supplementation successfully raises vitamin D levels above the 30 ng/mL threshold.

Question 17: Proportion CI in R (5 points)

Which R function would you use to calculate a confidence interval for a population proportion?

- a) `prop.test(x, n)`
- b) `t.test(x)`

- c) `binom.test(x, n)`
- d) Both a and c are appropriate